INSTRUCTOR GUIDE & ANSWER KEY

SPHERE: MISSION CONTROL

INSTRUCTOR MANUAL, SAFETY GUIDANCE AND ASSESSMENT KEY

Scope: The activity is intended for introductory undergraduate engineering or physics teaching. Instructors should adapt the risk assessment, supervision ratio and assessment weighting to local institutional requirements.

1. LEARNING OUTCOMES

On completion, students should be able to:

- Distinguish conduction, convection and radiation in the test system.
- Apply Newton's law of cooling,

$$T(t) = T_{\text{env}} + (T_0 - T_{\text{env}})e^{-kt},$$

  and interpret the fitted cooling constant k (min^{-1}).
- Acquire and plot temperature data (T_{obs}) at defined time intervals ($t = 0 \text{ to } 15 \text{ min}$).
- Calculate residuals ($|\Delta T| = |T_{\text{obs}} - T_{\text{model}}|$) and MAE ($\text{MAE} = \frac{1}{n}\sum_{i=1}^{n}|\Delta T_i|$), then discuss uncertainty and limitations of the lumped-parameter model.

2. 60-MINUTE LESSON ROADMAP

TIMELINE	LESSON PHASE	WHAT YOU & YOUR STUDENTS ARE DOING
$0–10\ \mathrm{min}$	1. Theory and concept check	Review the three heat-transfer modes, the water-bath boundary condition ($T_{\mathrm{env}} = 0.0^{\circ}\mathrm{C}$) and the limits of the spaceflight analogy. Complete the preliminary concept check.
$10–20\ \mathrm{min}$	2. Capsule Assembly	Assign teams their insulation setup (A: bare, B: bubble wrap, C: reflective film, D: multilayer assembly). Demonstrate probe placement and the pre-immersion leak test.
$20–35\ \mathrm{min}$	3. Fifteen-minute test	Add $100\ \mathrm{mL}$ of water at $80.0^{\circ}\mathrm{C}$ to each capsule. Measure the bath temperature, immerse the capsules, start the timer and record temperatures every $60\ \mathrm{seconds}$.
$35–48\ \mathrm{min}$	4. Analysis	Plot measured (T_{obs}) and predicted (T_{model}) temperatures, calculate residuals and MAE ($\mathrm{MAE} = \frac{1}{n}\sum \lvert\Delta T_i\rvert$), and compare the error with instrument accuracy ($\pm 0.5^{\circ}\mathrm{C}$).
$48–60\ \mathrm{min}$	5. Technical discussion	Compare the four configurations and discuss material mechanisms, model assumptions, uncertainty and design trade-offs.

3. LAB SETUP & SAFETY CHECKLIST

Complete and document the following controls before the session:

- **Hot-water control:** Do not use boiling water. Prepare water at $80.0^{\circ}\mathrm{C}$ and use heat-resistant gloves, eye protection and a stable pouring area in accordance with the local risk assessment.
- **Bath condition:** Prepare an ice-water slurry and measure its temperature. Do not assume an exact $0.0^{\circ}\mathrm{C}$ boundary without verifying.
- **Leak Prevention:** Check that students seal the probe opening using the approved lab method and complete a leak test before immersion.
- **Probe Placement:** Check that each probe hangs in the center of the liquid and isn't pressed against the plastic jar walls.

4. REFERENCE-MODEL OUTPUT

The table lists outputs from the assumed coefficients at $t = 15\ \mathrm{min}$, with $T_0 = 80.0^\circ\mathrm{C}$ and $T_{\mathrm{env}} = 0.0^\circ\mathrm{C}$. These are model values (T_{model}), not acceptance criteria for measured results.

INSULATION TYPE	ASSUMED COOLING CONSTANT (k, min^{-1})	MODEL TEMP AT 15 MIN	QUALITATIVE INTERPRETATION
A. Bare Control	$k \approx 0.150\ \mathrm{min}^{-1}$	$8.0^\circ\mathrm{C}$–$12.0^\circ\mathrm{C}$	Rapid heat loss through unshielded conduction into water and internal liquid circulation.
B. Bubble Wrap	$k \approx 0.040\ \mathrm{min}^{-1}$	$\approx 43.9^\circ\mathrm{C}$	Sealed air cells reduce fluid motion and slow heat transfer.
C. Mylar Foil	$k \approx 0.143\ \mathrm{min}^{-1}$	$\approx 9.4^\circ\mathrm{C}$	Reflective film is highly sensitive to conductive contact and leakage in the water-bath setup.
D. Multilayer assembly	$k \approx 0.015\ \mathrm{min}^{-1}$	$\approx 63.9^\circ\mathrm{C}$	The combined layers increase effective thermal resistance through trapped air, reduced fluid motion and lower radiative exchange.

Optional extension: A cotton-only configuration may be added as a fifth trial when time permits.

SECTION 5: OFFICIAL SOLUTIONS & TEACHER ANSWER KEY

ANSWER KEY

Question 1: Why doesn't Mylar foil work by itself in cold water?

Sample answer: A reflective film can reduce radiative exchange, but it has little thickness and therefore provides limited resistance to conduction when it is in direct contact with the capsule and surrounding water. A low-conductivity spacer introduces trapped air and additional thermal resistance.

Marking guidance: award credit for distinguishing radiative performance from conductive resistance.

Question 2: What does the cooling constant (k) tell us?

Sample answer: Within the same test geometry and boundary conditions, a smaller fitted k means a slower exponential approach to T_{env} ($T(t) = T_{\mathrm{env}} + (T_0 - T_{\mathrm{env}})e^{-kt}$). The value is an effective system parameter with units of min^{-1}; it is not the material thermal conductivity λ.

Question 3: Why can the multilayer assembly reduce cooling?

Sample answer: The materials introduce complementary resistance mechanisms:

- **Cotton** traps air and reduces effective conduction.
- **Bubble wrap** limits air movement within sealed cells.
- **Reflective film** reduces radiative exchange when an air gap is present.

Interfaces between layers also add contact resistance. The measured result must be used to determine the actual performance ($k \approx 0.015\ \mathrm{min}^{-1}$); no specific final temperature should be assumed without empirical data.

Question 4: Where could experimental errors come from?

Sample Answer (Accept any two sensible answers):

1. The sensor tip was touching the cold plastic jar wall instead of hanging in the middle of the water.
2. Water leaked into the jar through an incompletely sealed lid or probe opening.
3. The ice bath warmed up above $0.0^\circ\mathrm{C}$ during the 15 minutes.
4. The water inside the jar wasn't stirred, so warm water rose to the top and cold water sank to the bottom.

Question 5: What happens if your ice bath warms up?

Sample Answer: Students should measure the real bath temperature (e.g. $T_{\mathrm{env}} = 4.5^\circ\mathrm{C}$) and type that new number into the T_{env} box in the Mission Control simulator. Updating the environment temperature makes the theoretical curve ($T(t) = T_{\mathrm{env}} + (T_0 - T_{\mathrm{env}})e^{-kt}$) match the real classroom conditions.
Grading Tip: Award credit if they mention updating T_{env} or the environment temperature input in the software.

Question 6: What if we double the water volume?

Sample Answer: Doubling the water from $100\ \mathrm{mL}$ to $200\ \mathrm{mL}$ doubles the mass of hot liquid ($Q = m \cdot c \cdot \Delta T$), meaning it holds twice as much thermal energy. Because there is more heat to lose through roughly the same surface area, it cools down **slower**, and the cooling constant k **decreases**.
Grading Tip: Look for: cools slower and k decreases.

Question 7: Spacesuit Design Trade-Offs

Sample Answer: You can't just add 50 layers because the spacesuit would become too heavy, bulky, and stiff. Astronauts wouldn't be able to bend their elbows or fingers during a 7-hour spacewalk. Also, human bodies generate lots of body heat when working hard - too much insulation would trap that heat and cause dangerous overheating without heavy water-cooling systems.
Grading Tip: Accept answers discussing joint flexibility/mobility, suit weight/bulk, or body heat buildup.

Question 8: How can the MAE be reduced?

Sample Answer:

1. Use a consistent lid-sealing method and leak-test the capsule before immersion.
2. Position the probe consistently near the centre of the liquid without contacting the wall ($\pm 0.5^\circ\mathrm{C}$).
3. Swirl the jar gently before reading to keep the water mixed evenly.
4. Take readings at consistent recorded times ($\mathrm{MAE} = \frac{1}{n}\sum_{i=1}^{n} |T_{\mathrm{obs},i} - T_{\mathrm{model},i}|$).

6. GRADING RUBRIC & CHECKLIST

CRITERIA	EXCELLENT (90-100%)	GOOD (75-89%)	NEEDS WORK (<75%)
Data Logging	All 15 minutes filled with accurate readings, absolute differences, and helpful notes.	Completed with only 1-2 missed minutes; minor calculation mistakes.	Multiple blank rows; missing calculations.
Graphing	Clean, accurate points plotted with a smooth line; dashed theoretical curve clearly shown.	Points plotted with minor curve scaling inaccuracies.	Points not connected; axes mislabeled or missing lines.
Analysis Questions	Clear, thoughtful answers explaining conduction, convection, radiation, and MLI synergy (Q1-Q8).	Good understanding of heat transfer with minor details missing.	Confuses conduction with radiation; incomplete answers.